MatterGen: a generative model for inorganic materials design

Claudio Zeni[1†], Robert Pinsler[1†], Daniel Zügner[1†], Andrew Fowler[1†], Matthew Horton[1†], Xiang Fu[1], Aliaksandra Shysheya[1], Jonathan Crabbé[1], Lixin Sun[1], Jake Smith[1], Bichlien Nguyen[1], Hannes Schulz[1], Sarah Lewis[1], Chin-Wei Huang[1], Ziheng Lu[1], Yichi Zhou[1], Han Yang[1], Hongxia Hao[1], Jielan Li[1], Ryota Tomioka[1*†], Tian Xie[1*†]

[1]Microsoft Research AI4Science.

*Corresponding author(s). E-mail(s): ryoto@microsoft.com; tianxie@microsoft.com;
†Equal contribution; non-corresponding authors are listed in random order.

Abstract

The design of functional materials with desired properties is essential in driving technological advances in areas like energy storage, catalysis, and carbon capture [1–3]. Generative models provide a new paradigm for materials design by directly generating entirely novel materials given desired property constraints. Despite recent progress, current generative models have low success rate in proposing stable crystals, or can only satisfy a very limited set of property constraints [4–13]. Here, we present MatterGen, a model that generates stable, diverse inorganic materials across the periodic table and can further be fine-tuned to steer the generation towards a broad range of property constraints. To enable this, we introduce a new diffusion-based generative process that produces crystalline structures by gradually refining atom types, coordinates, and the periodic lattice. We further introduce adapter modules to enable fine-tuning towards any given property constraints with a labeled dataset. Compared to prior generative models, structures produced by MatterGen are more than twice as likely to be novel and stable, and more than 15 times closer to the local energy minimum. After fine-tuning, MatterGen successfully generates stable, novel materials with desired chemistry, symmetry, as well as mechanical, electronic and magnetic properties.

Finally, we demonstrate multi-property materials design capabilities by proposing structures that have both high magnetic density and a chemical composition with low supply-chain risk. We believe that the quality of generated materials and the breadth of MatterGen's capabilities represent a major advancement towards creating a universal generative model for materials design.

1 Introduction

The rate at which we can discover better materials has a major impact on the pace of technological innovation in areas such as carbon capture, semiconductor design, and energy storage. Traditionally, most novel materials have been found through experimentation and human intuition, which require long iteration cycles and are limited by the number of candidates that can be tested. Thanks to the advance of high throughput screening [14], open material databases [15–19], machine-learning-based property predictors [20, 21], and machine learning force fields (MLFFs) [22, 23], it has become increasingly common to screen hundreds of thousands of materials to identify promising candidates [24–27]. However, screening-based methods are still fundamentally limited by the number of known materials. The largest explorations of previously unknown crystalline materials are in the orders of 10^6–10^7 materials [23, 27–29], which is only a tiny fraction of the number of potential stable inorganic compounds (10^{10} quaternary compounds only considering stoichiometry [30]). Moreover, these methods cannot be efficiently steered towards finding materials with target properties.

Given these limitations, there has been an enormous interest in the inverse design of materials [31–34]. The aim of inverse design is to directly generate material structures that satisfy possibly rare or even conflicting target property constraints, e.g., via generative models [4, 9, 13], evolutionary algorithms [35], and reinforcement learning [36]. Generative models are particularly promising since they have the potential to efficiently explore entirely new structures, yet they can also be flexibly adapted to different downstream tasks. Despite recent progress, current generative models often fall short of producing stable materials according to density functional theory (DFT) calculations [4, 5, 37, 38], are constrained by a narrow subset of elements [8, 10, 11], and/or can only optimize a very limited set of properties, mainly formation energy [4, 5, 9, 13, 38–40].

In this study, we present MatterGen, a diffusion-based generative model for designing stable inorganic materials across the periodic table. MatterGen can further be fine-tuned via adapter modules to steer the generation towards materials with desired chemical composition, symmetry, and scalar property (e.g., band gap, bulk modulus, magnetic density) constraints. Compared to the previous state-of-the-art generative model for materials [4], MatterGen more than doubles the percentage of generated stable, unique, and novel (S.U.N.) materials, and generates structures that are more than 15 times closer to their ground-truth structures at the DFT local energy minimum. When fine-tuned, MatterGen often generates more S.U.N. materials in target chemical systems than well-established methods like substitution and random structure search (RSS), is capable of generating highly symmetric structures given the desired

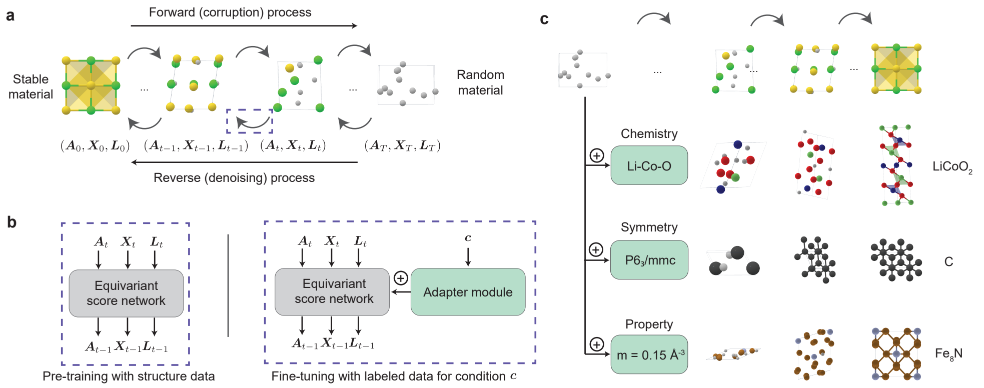

Fig. 1: **Inorganic materials design with MatterGen.** **(a)** MatterGen generates stable materials by reversing a corruption process through iteratively denoising an initially random structure. The forward diffusion process is designed to independently corrupt atom types $\boldsymbol{A}$, coordinates $\boldsymbol{X}$, and the lattice $\boldsymbol{L}$ to approach a physically motivated distribution of random materials. **(b)** An equivariant score network is pre-trained on a large dataset of stable material structures to jointly denoise atom types, coordinates, and the lattice. The score network is then fine-tuned with a labeled dataset through an adapter module that alters the model using the encoded property $\boldsymbol{c}$. **(c)** MatterGen can be fine-tuned to steer the generation towards materials with desired chemistry, symmetry, and scalar property constraints.

space groups, and directly generates S.U.N. materials that satisfy target mechanical, electronic, and magnetic property constraints. Finally, we showcase MatterGen's ability to design materials given multiple property constraints by generating promising materials that have both high magnetic density and a chemical composition with low supply-chain risk.

2 Results

2.1 Diffusion process for crystalline material generation

In MatterGen, we introduce a novel diffusion process tailored for crystalline materials (Fig. 1(a)). Diffusion models generate samples by learning a score network to reverse a fixed corruption process [41–43]. Corruption processes for images typically add Gaussian noise but crystalline materials have unique periodic structures and symmetries which demand a customized diffusion process. A crystalline material can be defined by its repeating unit, i.e., its unit cell, which encodes the atom types $\boldsymbol{A}$ (i.e., chemical elements), coordinates $\boldsymbol{X}$, and periodic lattice $\boldsymbol{L}$ (Appendices A.1 and A.2). We define a corruption process for each component that suits their own geometry and has physically motivated limiting noise distributions. More concretely, the coordinate diffusion respects the periodic boundary by employing a wrapped Normal distribution and approaches a uniform distribution at the noisy limit (Appendix A.6). The lattice diffusion takes a symmetric form and approaches a distribution whose mean is a cubic lattice with average atomic density from the training data (Appendix A.7). The atom diffusion is defined in categorical space where individual atoms are corrupted into a masked state (Appendix A.5). Given the corrupted structure, we learn a score network that outputs equivariant scores for atom type, coordinate, and lattice, respectively, which removes the need to learn the symmetries from data (Appendix A.8). We refer to this network as the base model.

To generate materials with desired property constraints, we introduce adapter modules that can be used for fine-tuning the base model on an additional dataset with property labels (Fig. 1(b), more details in Appendix B). Fine-tuning is particularly appealing as it still works well if the labeled dataset is small compared to unlabeled structure datasets, which is often the case due to the high computational cost of calculating properties. The adapter modules are tunable components injected into each layer of the base model to alter its output depending on the given property label [44]. The resulting fine-tuned model is used in combination with classifier-free guidance [45] to steer the generation towards target property constraints. We apply this approach to multiple types of properties, producing a set of fine-tuned models that can generate materials with target chemical composition, symmetry, or scalar properties such as magnetic density (Fig. 1(c)).

2.2 Generating stable, diverse materials

We formulate learning a generative model for inverse materials design as a two-step process, where we first pre-train a general base model for generating stable, diverse crystals across the periodic table, and then we fine-tune this base model towards different downstream tasks. In this section, we focus on the ability of MatterGen's base model to generate stable, diverse materials, which we argue is a prerequisite for addressing any inverse materials design task. Since diversity is difficult to measure directly, we resort to quantifying MatterGen's ability to generate S.U.N. materials, and provide an additional analysis of the quality and diversity of generated structures. To train the base model, we curate a large, diverse dataset comprising 607,684 stable structures with up to 20 atoms recomputed from the Materials Project (MP) [15] and Alexandria [29, 46] datasets, which we refer to as Alex-MP-20. We consider a structure to be stable if its energy per atom after relaxation via DFT is below the 0.1 eV/atom threshold of a reference dataset comprising 1,081,850 unique structures recomputed from the MP, Alexandria, and Inorganic Crystal Structure Database (ICSD) datasets. We refer to this dataset as Alex-MP-ICSD. We consider a structure to be novel if it is not contained in Alex-MP-ICSD. We adopt these definitions throughout unless stated otherwise. More details are in Appendices C and D.3.1.

Fig. 2(a) shows several random samples generated by MatterGen, featuring typical coordination environments of inorganic materials; see Appendix D.3.2 for a more detailed analysis. To assess stability, we perform DFT calculations on 1024 generated structures. Fig. 2(b) shows that 78 % of generated structures fall below the 0.1 eV/atom threshold (13 % below 0.0 eV/atom) of MP's convex hull, while 75 % fall below the 0.1 eV/atom threshold (3 % below 0.0 eV/atom) of the combined Alex-MP-ICSD hull (Fig. 2(b)). Further, 95 % of generated structures have an RMSD w.r.t.

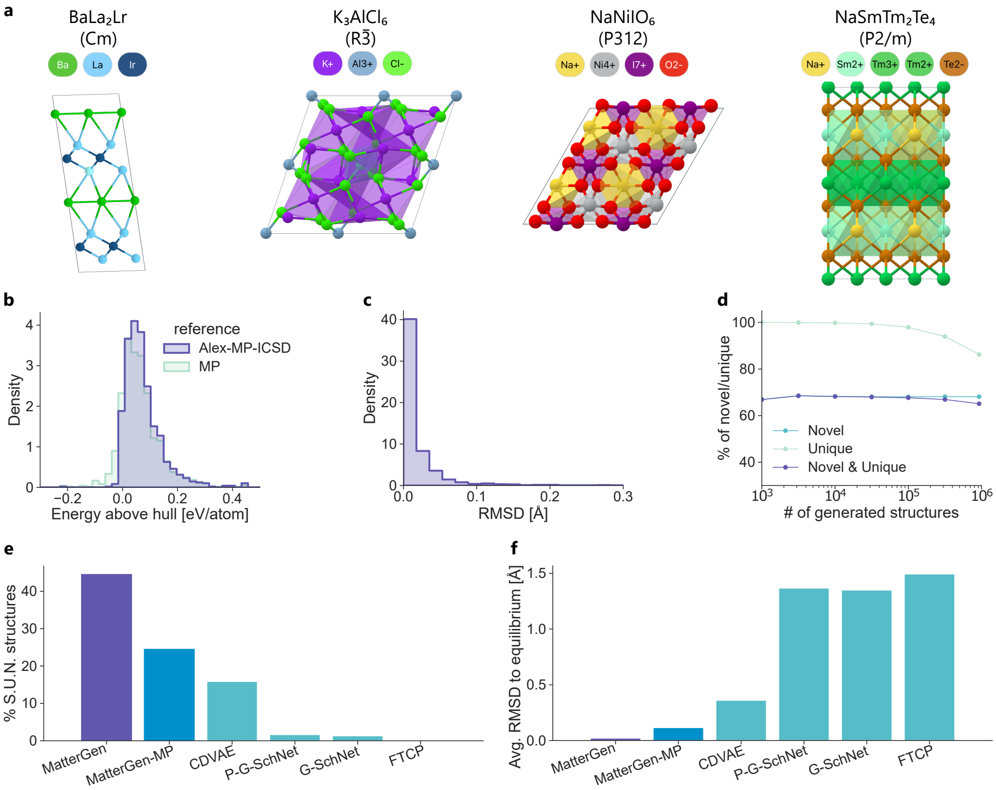

Fig. 2: **Generating stable, unique and novel inorganic materials. (a)** Visualization of four randomly selected crystals generated by MatterGen, with corresponding chemical formula and space group symbols. **(b)** Distribution of energy above the hull using MP and Alex-MP-ICSD dataset as energy references, respectively. **(c)** Distribution of root mean squared displacement (RMSD) between initial generated structures and DFT relaxed structures. **(d)** Percentage of unique, novel structures as a function of number of generated structures. Novelty is defined with respect to Alex-MP-ICSD. **(e-f)** Percentage of S.U.N. structures (e) and average RMSD between initial and DFT-relaxed structures (f) for MatterGen, MatterGen-MP, and several baseline models, including CDVAE [4], P-G-SchNet, G-SchNet [47], and FTCP [38].

their DFT-relaxed structures that is below 0.076 Å (Fig. 2(c)), which is almost one order of magnitude smaller than the atomic radius of the hydrogen atom (0.53 Å). These results indicate that the majority of structures generated by MatterGen are stable, and very close to the DFT local energy minimum. We further investigate whether MatterGen can generate a substantial amount of unique and novel materials. We showcase in Fig. 2(d) that the percentage of unique structures is 100 % when generating 1000 structures and only drops to 86 % after generating one million structures, while novelty remains stable around 68 %. This suggests that MatterGen is able to generate